

Transformers — Solution

X26 (2026) — English translation

A1

0.20 pts

What are the magnetic field fluxes through the primary and secondary windings Φ_1 and Φ_2 ? Express your answers using Φ , N_1 and N_2 .

Answer: \textbf{Answer:}

$$\Phi_1 = \Phi \cdot N_1, \quad \Phi_2 = -\Phi \cdot N_2$$

A2

0.20 pts

Using the circulation theorem for the magnetic field strength vector $\vec{H}$, find the magnetic field flux Φ in the core. Express your answer in terms of S , ℓ , μ , I_1 , I_2 , N_1 and N_2 .

Let us write the circulation theorem for the entire transformer:

$$H \cdot \ell = N_1 I_1 - N_2 I_2,$$

Let us express the magnetic field flux:

Answer: \textbf{Answer:}

$$\Phi = BS = \mu\mu_0 HS = \frac{\mu\mu_0 S}{\ell} (N_1 I_1 - N_2 I_2)$$

A3

0.30 pts

Obtain expressions for the inductances of windings L_1 , L_2 and their mutual inductance coefficient M . Express your answer using S , ℓ , μ , I_1 , I_2 , N_1 and N_2 .

Answer: \textbf{Answer:}

$$\begin{aligned} L_1 &= \frac{\partial \Phi_1}{\partial I_1} = \frac{\mu\mu_0 S N_1^2}{\ell} \\ L_2 &= \frac{\partial \Phi_2}{\partial I_2} = \frac{\mu\mu_0 S N_2^2}{\ell} \\ M &= \frac{\partial \Phi_1}{\partial I_2} = \frac{\partial \Phi_2}{\partial I_1} = -\frac{\mu\mu_0 S N_1 N_2}{\ell} \end{aligned}$$

A4

0.40 pts

Find the magnetic induction emf $\mathcal{E}_1$ and $\mathcal{E}_2$ arising in the primary and secondary windings, respectively. Express your answer using $\dot{I}_1$, $\dot{I}_2$, L_1 , L_2 and M . Notes: EMF $\mathcal{E}_1$ is considered positive if the electrostatic potential at the top point $a1$ is greater than at the bottom $a2$. EMF $\mathcal{E}_2$ is considered positive if the electrostatic potential at the top point $b1$ is greater than at the bottom $b2$.

Find the magnetic induction emf $\mathcal{E}_1$ and $\mathcal{E}_2$ arising in the primary and secondary windings, respectively. Express your answer using $\dot{I}_1$, $\dot{I}_2$, L_1 , L_2 and M .

Notes:

Answer: \textbf{Answer:}

$$\begin{aligned}\mathcal{E}_1 &= \dot{\Phi}_1 = N_1 \dot{\Phi} = L_1 \dot{I}_1 + M \dot{I}_2 \\ \mathcal{E}_2 &= -\dot{\Phi}_2 = N_2 \dot{\Phi} = -L_2 \dot{I}_2 - M \dot{I}_1\end{aligned}$$

A5

0.50 pts

Obtain an expression for the current transformation ratio $k_I \equiv \left| \tilde{I}_1 \right| / \left| \tilde{I}_2 \right|$. The answer may include L_1 , L_2 , ω and Z .

Using the equations from A4 and the following relations:

$$U_2 = I_2 Z M^2 = L_1 L_2$$

$$I_2 Z = \mathcal{E}_2 = -L_2 \dot{I}_2 - M \dot{I}_1 I_2 (Z - i\omega L_2) = i\omega M I_1$$

we obtain

Answer: \textbf{Answer:}

$$k_I = \left| \frac{\tilde{I}_1}{\tilde{I}_2} \right| = \left| \frac{Z - i\omega L_2}{i\omega M} \right| = \frac{\sqrt{(\operatorname{Re} Z)^2 + (\operatorname{Im} Z - \omega L_2)^2}}{\omega \sqrt{L_1 L_2}}$$

A6

0.50 pts

What is the voltage transformation ratio $k_U = \left| \tilde{U}_2 \right| / \left| \tilde{U}_1 \right|$? The answer may include L_1 , L_2 , ω and Z .

Writing down the equations from the previous paragraphs:

$$U_2 = -L_2 \dot{I}_2 - M \dot{I}_1 U_2 = I_2 \cdot Z U_1 = L_1 \dot{I}_1 + M \dot{I}_2$$

we express U_2 via/through U_1 :

$$U_2 = i\omega L_2 \frac{U_2}{Z} + i\omega M I_1 U_1 = -i\omega L_1 I_1 - i\omega M \frac{U_2}{Z} U_2 \left(1 - \frac{i\omega L_2}{Z} \right) = \frac{M}{L_1} \left(U_1 - \frac{i\omega M}{Z} U_2 \right) \frac{U_2}{U_1} = \frac{\frac{M}{L_1}}{1 - \frac{i\omega L_2}{Z} + \frac{i\omega M^2}{Z L_1}} = \frac{M}{L_1} = -$$

we get:

Answer: \textbf{Answer:}

$$k_U = \left| -\frac{N_2}{N_1} \right| = \sqrt{\frac{L_2}{L_1}}$$

A7

0.30 pts

Write down the condition for the number of turns in windings N_1 and N_2 , under which the expression (1) is satisfied.

Answer: \textbf{Answer:}

$$N_2 \gg N_1$$

A8

0.60 pts

Determine the phase difference $\Delta\varphi$ by which voltage U_1 leads current I_1 . The answer may include L_1 , L_2 , ω and Z .

$$U_1 + \frac{i\omega M}{Z} \cdot U_2 = -i\omega L_1 I_1$$

$$U_1 \left(1 - \frac{i\omega M N_2}{Z N_1} \right) = -i\omega L_1 I_1$$

$$U_1 = \frac{-i\omega L_1 Z N_1}{Z N_1 - i\omega M N_2} I_1$$

after transformations:

Or, which is convenient for further calculations,

$$\cos \Delta\varphi = \frac{\omega L_2 \operatorname{Re} Z}{|Z| \sqrt{(\operatorname{Re} Z)^2 + (\omega L_2 - \operatorname{Im} Z)^2}}$$

Answer: \textbf{Answer:}

$$\operatorname{tg} \Delta\varphi = \frac{|Z|^2 - \omega L_2 \cdot \operatorname{Im} Z}{\omega L_2 \operatorname{Re} Z}$$

A9

0.60 pts

Determine the time-averaged power P of source U_1 . The answer may include U_0 , Z , N_1 and N_2 .

The average power is found as:

$$P = \frac{1}{2} |I_1| |U_1| \cos \Delta\varphi$$

Substituting we get:

The calculations could be significantly reduced by using the following formula for power:

$$P = \operatorname{Re} \left(\frac{1}{2} I_1 U_1^* \right)$$

Answer: \textbf{Answer:}

$$P = \frac{U_1^2 \operatorname{Re} Z}{2|Z|^2} \cdot \frac{N_2^2}{N_1^2}$$

B1

0.50 pts

Find the ratio I/I_A .

Using the formula for k_I we get the answer:

Answer: \textbf{Answer:}

$$\frac{I}{I_A} = \sqrt{\frac{L_1}{L_2}} = N$$

C1

0.20 pts

Obtain an expression for the average power P_0 , which is released at the load if the winding resistance can be neglected. Express your answer through the amplitude of the input voltage U_0 and the load resistance R . Calculate it for a transformer whose parameters were given in the problem earlier.

Using the formula for average power:

Answer: \textbf{Answer:}

$$P_0 = \frac{U_0^2}{2R} = \frac{U_{\text{rms}}^2}{R} = 303 \text{ W}$$